

Assignment-Based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?
 - **Seasonality:** Bike demand peaks heavily during Fall, followed closely by Summer. Demand drops drastically during Spring due to colder temperatures.
 - **Weather Conditions:** Clear or partly cloudy days (weathersit_Clear) show a massive positive effect on bike rentals. Conversely, days with light snow or rain show a profound negative impact, while heavy rain/ice halts demand entirely.
 - **Months & Days:** Demand shows an upward trajectory during the mid-year months (June through September). Weekdays vs. weekends show minimal structural variance, indicating steady, consistent daily utility.
2. Why is it important to use drop_first=True during dummy variable creation?
 - When converting a categorical variable with n distinct categories into dummy variables, we only need $n-1$ binary columns to perfectly capture all the information. If we keep all n columns, the final column can be perfectly predicted by the combinations of the others. This creates perfect multicollinearity—known as the **Dummy Variable Trap**—which makes the data matrix singular and prevents the linear regression algorithm from reliably calculating stable coefficients.
3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?
 - Looking at the numerical features (temp, atemp, hum, windspeed) against the target variable (cnt), the temperature variables (**temp** and **atemp**) display the steepest linear upward trend, indicating they possess the **highest positive correlation** with bike rental demand.
4. How did you validate the assumptions of Linear Regression after building the model on the training set?

After building the model on the training set, the core assumptions were validated using these diagnostics:

- **Linearity:** Inspected scatter plots of independent variables against cnt to confirm stable linear patterns.
- **Normality of Residuals:** Plotted a histogram/KDE of the error terms ($y_{\text{train}} - y_{\text{pred}}$). The distribution formed a perfectly symmetrical, zero-centered bell curve, confirming the residuals are normally distributed.
- **Homoscedasticity:** Plotted the residuals against the predicted values to verify that the variance of the errors remains entirely constant across the prediction spectrum (no funnel shapes).

- **Independence (No Multicollinearity):** Calculated the Variance Inflation Factor (VIF) for all chosen features, ensuring every final feature scored well below the threshold of 5.
5. **Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?**
- Looking at the absolute magnitude of the final regression coefficients (coef) in our optimized model, the top 3 features driving bike demand are:
 - **weathersit_Light_Snow_Rain (Coefficient: -0.3207):** This is the strongest negative predictor. Bad weather (light snow, rain, or thunderstorms) causes a massive structural drop in rental demand.
 - **yr (Coefficient: +0.2457):** A powerful positive predictor indicating that, holding all other factors constant, there is a substantial organic increase in bike demand from 2018 to 2019.
 - **season_spring (Coefficient: -0.2381):** A heavy negative predictor showing that rental bookings drop significantly during the cold spring months compared to the baseline season.

General Subjective Questions:

1. Explain the linear regression algorithm in detail:

Linear Regression is a supervised machine learning algorithm used to model the linear relationship between a continuous dependent target variable (Y) and one or more independent predictor variables (X). The mathematical equation is written as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_n x_n + \epsilon$$

Where β_0 is the y-intercept, $\beta_1, \dots, \beta_n$ are the feature coefficients (weights), and ϵ represents the random error term. The algorithm utilizes the **Ordinary Least Squares (OLS)** method to estimate the weights. OLS systematically minimizes the Residual Sum of Squares (RSS)—which is the sum of the squared differences between the actual data points (y_i) and the predicted regression line ($\hat{y}$):

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

2. Explain the Anscombe's quartet in detail.

Anscombe's Quartet comprises four distinct datasets constructed by statistician Francis Anscombe in 1973. Crucially, all four datasets share identical numerical statistical properties (same mean, variance, correlation coefficient, and regression line equation).

Despite having the exact same statistical summaries, when visualized on a scatter plot, they reveal completely different geometric configurations. It proves that relying strictly on raw numerical summaries can be highly misleading, emphasizing that **visual data exploration** is a mandatory prerequisite before running any algorithm.

3. What is Pearson's R?

Pearson's Correlation Coefficient (r) is a metric that measures the linear strength and direction between two continuous variables. The value is always constrained strictly between -1.0 and +1.0:

$r = +1$: Perfect positive linear relationship.

$r = -1$: Perfect negative linear relationship.

$r = 0$: Absolutely no linear correlation between the variables.

Mathematically, it is calculated by dividing the covariance of the two variables by the product of their individual standard deviations:

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Scaling is a data preprocessing technique used to bring all numerical features down to a matching, uniform scale without altering the underlying shapes of their distributions. It is vital because features with naturally massive numeric ranges will mathematically dominate the model's cost functions incorrectly.

Normalized Scaling (MinMax Scaling): Shifts and rescales data precisely between 0 and 1 using the formula $X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$ It is highly sensitive to outliers.

Standardized Scaling (Z-score Normalization): Centers the data around a mean of 0 with a standard deviation of 1 using the formula $X_{std} = \frac{X - \mu}{\sigma}$. It is much more robust to outliers.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

The Variance Inflation Factor (VIF) is calculated using the formula:

$$VIF = \frac{1}{1 - R_i^2}$$

Where R_i^2 is the R-squared score obtained by regressing that specific feature against all other independent variables. If two or more independent variables possess perfect multicollinearity (meaning one variable can be 100% perfectly predicted by another), the R_i^2 value becomes exactly 1. Plugging this into the formula yields $\frac{1}{1-1} = \frac{1}{0}$, which results in an mathematical value of infinity.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

A Quantile-Quantile (Q-Q) plot is a graphical diagnostic tool used to determine if a set of data points follows a specific theoretical distribution—most commonly a standard normal distribution. It plots the empirical quantiles of your model's residuals against the expected mathematical quantiles of a perfect normal distribution. If the residuals are normal, they will align cleanly along a straight reference line. It provides an immediate, intuitive visual diagnostic check for the normality assumption.